9 Cryptography

9.4 Basic Facts About Modular Arithmetic

Example 9.4.1 Compute the following:

1. $47^8 \mod 49$
2. $65^5 \mod 35$

Example 9.4.2 Your turn! Compute the following:

1. $12^8 \mod 4$
2. $12^8 \mod 5$
3. $5^{12} \mod 4$
4. $98^{123} \mod 99$

What else can we do with modular arithmetic?

Example 9.4.3 Compute each of the following:

1. $3^{10} \mod 22$
2. $5^{10} \mod 22$
3. $21^{10} \mod 22$

Talk briefly about Euler's Theorem:

$$a^{\phi(n)} \equiv 1 \mod n$$

when $\gcd(a, n) = 1$.

Definition 9.4.4 Euler's totient function ϕ is defined as follows: If n is a positive whole number, and $p_1, p_2, \ldots, p_n$ are all the prime numbers dividing n, then

$$\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\cdots\left(1 - \frac{1}{p_n}\right)$$

Example 9.4.5 Compute the following:

1. $\phi(27)$
2. $\phi(19)$
3. $\phi(42)$
4. $\phi(124)$

16

Example 9.4.6 So how might we compute something like the following:

$$128^{1324} \mod 34$$

Example 9.4.7 Your turn! Compute

$$415^{1521} \mod 27.$$

1

9.4.1 Key Exchange

Up to this point every method of sending an encrypted message requires a secret key known to both Alice and Bob. A big problem in cryptography is the following:
Suppose that Alice and Bob are unable to meet in person, they have to email each other to agree on the secret key. Unfortunatly, Eve has access to Alice's emails, and thus can read any emails that are sent between Alice and Bob. How can Alice and Bob decide on a secret key without the key falling into Eve's hands?

We need some 'one-way function' that is easy to compute, but hard to undo. Illustrate how this can be done with the paint mixing example. In this example, the secret key is the final paint color.
Of course, since Alice and Bob are using computers, the one-way function will need to be a numerical function.